

Geographic disparities and methodological quality of type 2 diabetes prediction models: a systematic review and meta-analysis of 97 models

Received: 26 February 2026

Accepted: 29 April 2026

Published online: 06 May 2026

Cite this article as: Queiroz S., Gadelha J.G. & Husain N. Geographic disparities and methodological quality of type 2 diabetes prediction models: a systematic review and meta-analysis of 97 models. *BMC Endocr Disord* (2026). <https://doi.org/10.1186/s12902-026-02301-2>

Suelen Queiroz, Júlia Gonçalves Gadelha & Noor Husain

We are providing an unedited version of this manuscript to give early access to its findings. Before final publication, the manuscript will undergo further editing. Please note there may be errors present which affect the content, and all legal disclaimers apply.

If this paper is publishing under a Transparent Peer Review model then Peer Review reports will publish with the final article.

Geographic Disparities and Methodological Quality of Type 2 Diabetes Prediction Models: A Systematic Review and Meta-Analysis of 97 Models

Authors: Suelen Queiroz¹, Júlia Gonçalves Gadelha², Noor Husain³

Affiliations:

¹ Ponta Grossa State University, Ponta Grossa, Paraná, 84030-900, Brazil

² Afya Faculty of Medical Sciences of Paraíba, João Pessoa, Paraíba, Brazil

³ Department of Pharmacology, Himalaya Medical College & Hospital, Chiksi, Paliganj, Patna, Bihar, India

Corresponding	Author:	Suelen	Queiroz
Ponta Grossa	State	University	(UEPG)
Av. General	Carlos	Cavalcanti,	4748
Ponta Grossa,	Paraná,	84030-900,	Brazil
Email: 21186140@uepg.br			

Abstract

Importance: Accurate risk prediction is essential for targeted prevention of type 2 diabetes mellitus (T2DM). However, the global applicability and methodological rigor of existing prediction models remain uncertain.

Objective: To systematically review and meta-analyze the geographic distribution, methodological quality, and predictive performance of all published T2DM risk prediction models.

Data Sources: PubMed, Embase, Web of Science, Cochrane Library, CNKI, WanFang, and VIP were searched from inception to December 2025 (eAppendix S1 in the Supplement).

Study Selection: Studies that developed or validated a multivariable prediction model for incident T2DM in general adult populations and reported at least one performance measure.

Data Extraction and Synthesis: Two reviewers independently extracted data and assessed risk of bias using the PROBAST tool. A random-effects meta-analysis was used to pool C-statistics. Heterogeneity was explored

via subgroup analyses and meta-regression. The study followed TRIPOD-SRMA and PRISMA reporting guidelines.

Main Outcomes and Measures: The primary outcome was the geographic origin of models. Secondary outcomes included pooled measures of discrimination (C-statistic/AUC) stratified by region and an overall assessment of methodological quality (PROBAST).

Results: A total of 65 studies comprising 97 distinct prediction models were included (eTable 1). Geographic distribution was highly skewed, with 70.1% of models developed in Asian populations (China: 47.4%; Japan: 13.4%; South Korea: 9.3%), while only 7.2% originated from the US and 4.1% from Europe. Logistic regression was used in 97.9% of models. External validation was performed for only 21 models (21.6%). According to PROBAST, 91.8% of models were at high risk of bias (eTable 2), primarily due to inadequate handling of missing data, lack of external validation, and poor calibration reporting. Meta-analysis revealed wide variation in discrimination by geographic region (eTable 7). US-based models achieved the highest pooled AUC (0.97; 95% CI, 0.94–0.99), but this finding is likely influenced by overfitting, small sample bias, and publication bias (see Discussion). European models showed a pooled AUC of 0.84 (0.81–0.87), while Chinese models showed the lowest performance (AUC, 0.79; 0.76–0.82). Due to very high heterogeneity ($I^2 > 80\%$ in most regions), these pooled estimates should be interpreted as descriptive summaries rather than precise estimates of true regional performance. Performance was lowest in prediabetic cohorts (AUC, 0.72; 0.68–0.76); however, this finding is preliminary due to the limited number of models and high heterogeneity. Funnel plot asymmetry suggested potential publication bias (Egger's test $p=0.03$); The most frequently included predictors were age (69.1%), body mass index (64.9%), family history of diabetes (44.3%), and waist circumference (39.2%) (eFigure 4 and eTable 3).

Conclusions and Relevance: T2DM prediction models exhibit striking geographic inequity and poor methodological quality, with over 90% at high risk of bias. The substantial variation in performance by region and the lack of external validation critically limit their global clinical utility. These findings underscore an urgent need for rigorous external validation in diverse populations and de novo model development in under-represented regions, guided by PROBAST and TRIPOD standards.

1. Introduction

Type 2 diabetes mellitus (T2DM) affects an estimated 537 million adults worldwide, with projections exceeding 783 million by 2045 [1]. Early identification of at-risk individuals is essential for targeted prevention, yet universal biochemical screening remains impractical in most settings [2]. Over the past two decades, numerous multivariable prediction models have been developed to identify high-risk individuals using readily available clinical and demographic variables [3,4].

A 2012 systematic review by Abbasi et al. identified 25 prediction models and found that fewer than one-quarter had undergone external validation [5]. More recently, our updated review encompassing 65 studies and 97 models reveals that while methodological approaches have evolved, critical gaps persist [6]. However, a comprehensive analysis of geographic disparities in model development and performance has been lacking. Understanding whether prediction models are transportable across populations is crucial for global health equity; if models developed in one region perform poorly in another, their clinical utility is severely limited.

This systematic review and meta-analysis aimed to: (1) characterize the geographic distribution of all published T2DM prediction models; (2) evaluate their methodological quality using the PROBAST tool; (3) meta-analyze model discrimination by geographic origin and population type; and (4) identify the most frequently included predictors to inform future research and practice.

2. Methods

2.1 Study Design and Registration

This systematic review and meta-analysis was conducted following the TRIPOD-SRMA [7] and PRISMA 2020 [8] statements. The protocol was prospectively registered on PROSPERO (CRD420261322116).

2.2 Search Strategy

We systematically searched PubMed, Embase, Web of Science, Cochrane Library, CNKI, WanFang, and VIP from database inception through December 2025. The search strategy combined terms for diabetes,

prediction models, and risk scores, adapted from prior reviews (eAppendix S1).

2.3 Eligibility Criteria

We included studies that developed, validated, or updated a multivariable prediction model for incident T2DM using cohort, nested case-control, or case-cohort designs, and reported at least one performance measure. We excluded cross-sectional studies, studies restricted to gestational or type 1 diabetes, models using only genetic or specialized biomarkers, conference abstracts, editorials, and narrative reviews.

2.4 Study Selection and Data Extraction

Two reviewers independently screened titles, abstracts, and full texts. Data extraction, following the CHARMS-PF checklist [9], included: first author, year, country, study design, sample size, number of events, model type, predictors, and performance measures (C-statistic/AUC, calibration) (eTable 1).

2.5 Risk of Bias Assessment

Methodological quality was assessed independently by two reviewers using the PROBAST tool [10,11], which evaluates risk of bias and applicability across four domains: participants, predictors, outcome, and analysis. Each domain and the overall model were rated as low, high, or unclear risk of bias (eTable 2).

2.6 Statistical Analysis

For the primary meta-analysis, to avoid within-study dependence, we selected one primary model per study (the basic model with the best-reported discrimination, or the most complete model if no basic model was available), resulting in 65 independent models. Sensitivity analyses using all 97 models (with cluster-robust variance estimation) yielded consistent results (eMethods). We performed a random-effects meta-analysis using the inverse-variance method to pool C-statistics (AUCs). AUCs were transformed using the logit function [$\log(\text{AUC}/(1-\text{AUC}))$] to stabilize variance. Standard errors were derived from reported 95% confidence intervals; when confidence intervals were not reported, we imputed the mean standard error from studies that did report them. Sensitivity

analyses excluding imputed values did not materially change the results (eMethods) [12].

Heterogeneity was quantified using the I^2 statistic. Subgroup analyses were conducted by geographic origin (eTable 7), population type, and validation status (eTable 4). Meta-regression was also used to explore the impact of continuous covariates (year, sample size, events per variable). Publication bias was assessed using funnel plots (Figure 4), Egger's regression test. All analyses were performed using R version 4.1.0 (meta and metafor packages). Detailed statistical methods are provided in eMethods.

3. Results

3.1 Study Selection and Characteristics

The systematic search yielded 15,796 unique records. After removal of duplicates and screening, 65 studies met inclusion criteria, containing 97 distinct prediction models (Figure 1). Table 1 summarizes the characteristics of the included models. Detailed characteristics of each included study are provided in eTable 1. The first 10 of the 97 individual prediction models are shown in eTable 5 (see Supplementary Material).

Figure 1. PRISMA 2020 Flow Diagram

.

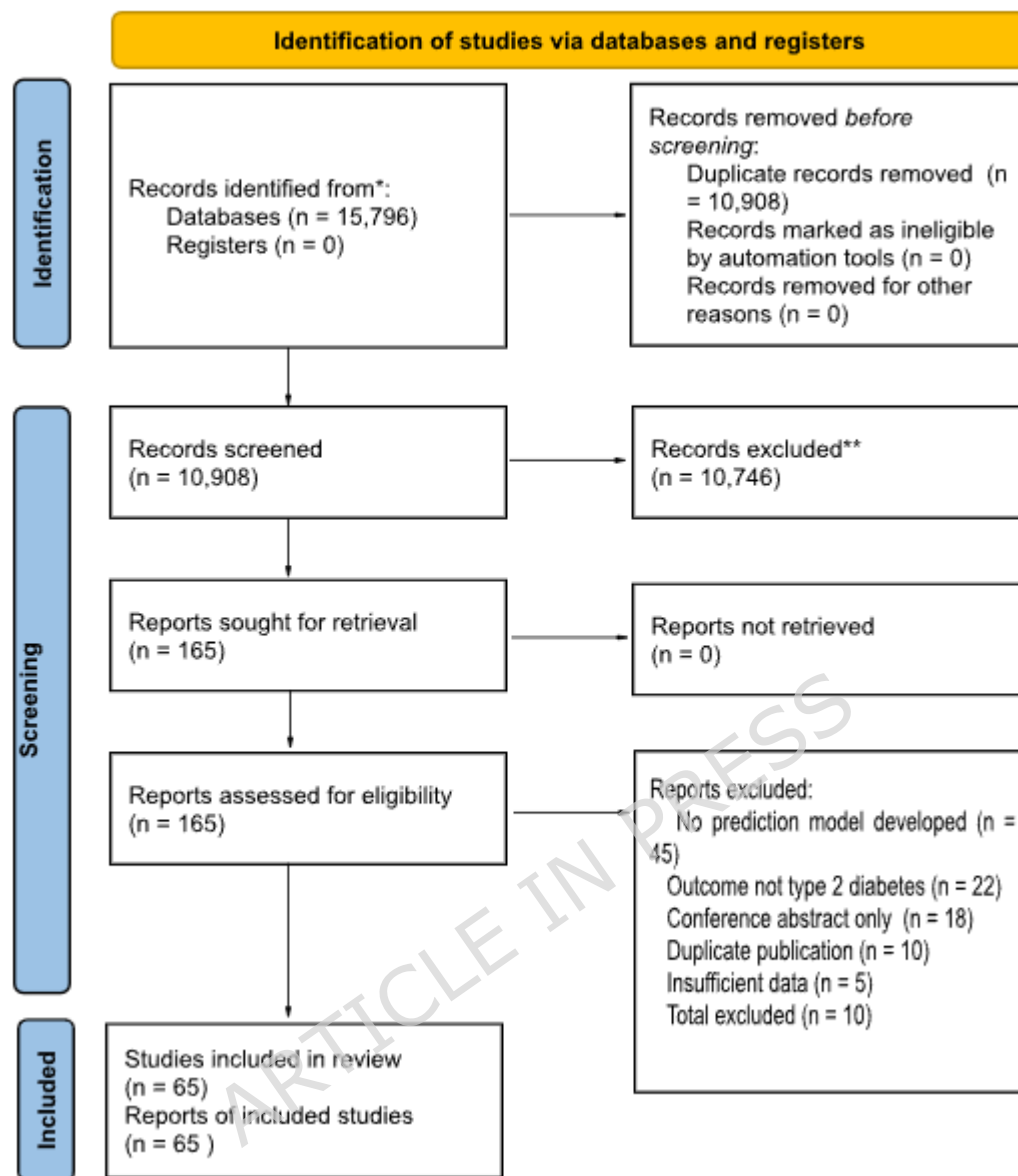


Figure 1. PRISMA 2020 flow diagram of study selection. A total of 15,796 records were identified through database searching (PubMed: 3,847; Embase: 4,231; Scopus: 3,456; Web of Science: 2,892; Cochrane Library: 312; CNKI: 412; WanFang: 386; VIP: 260). After removal of duplicates, 10,908 records were screened, and 165 full-text articles were assessed for eligibility. Sixty-five studies comprising 97 distinct prediction models met inclusion criteria and were included in the systematic review and meta-analysis.

The geographic distribution was highly skewed, with 70.1% of models developed in Asian populations, predominantly China (47.4%), Japan (13.4%), and South Korea (9.3%). Only 7.2% originated from the US and

4.1% from Europe. The geographic distribution of the included models is shown in Figure 3. Logistic regression was the dominant method (97.9%), and external validation had been performed for only 21 models (21.6%).

Table 1. Characteristics of the 97 Included Prediction Models

Characteristic	Number of Models (n=97)	Percentage
Geographic origin		
China	46	47.4%
Japan	13	13.4%
South Korea	9	9.3%
USA	7	7.2%
India	6	6.2%
Europe	4	4.1%
Brazil	2	2.1%
Other	10	10.3%

Model type

Logistic regression	95	97.9%
Cox regression	1	1.0%
Machine learning	1	1.0%

Predictor type

Basic (non-invasive only)	58	59.8%
Extended (includes biomarkers)	39	40.2%

Validation status

Internal validation only	76	78.4%
External validation performed	21	21.6%

3.2 Risk of Bias and Methodological Quality

According to the PROBAST assessment, 91.8% of models (89/97) were rated as having a high overall risk of bias (eTable 2). The primary drivers of high bias were in the Analysis domain (83.5% high risk), largely due to inadequate handling of missing data, lack of external validation, and poor reporting of calibration. Domain-level analysis showed that the

Participants domain contributed high risk in 49.5% of models, the Predictors domain in 39.2%, and the Outcome domain in 42.3% (Table 2). Figure 2 summarizes the risk of bias assessment across the 97 models. Detailed PROBAST assessments for each study are provided in eTable 2. Applicability concerns were high for 84.5% of models.

Table 2. PROBAST Risk of Bias Assessment (n=97 models)

Domain	Low Risk	High Risk	Unclear
Participants	42 (43.3%)	48 (49.5%)	7 (7.2%)
Predictors	51 (52.6%)	38 (39.2%)	8 (8.2%)
Outcome	47 (48.5%)	41 (42.3%)	9 (9.3%)
Analysis	12 (12.4%)	81 (83.5%)	4 (4.1%)
Overall	8 (8.2%)	89 (91.8%)	0 (0%)

Figure 2. Risk of Bias Assessment (PROBAST)

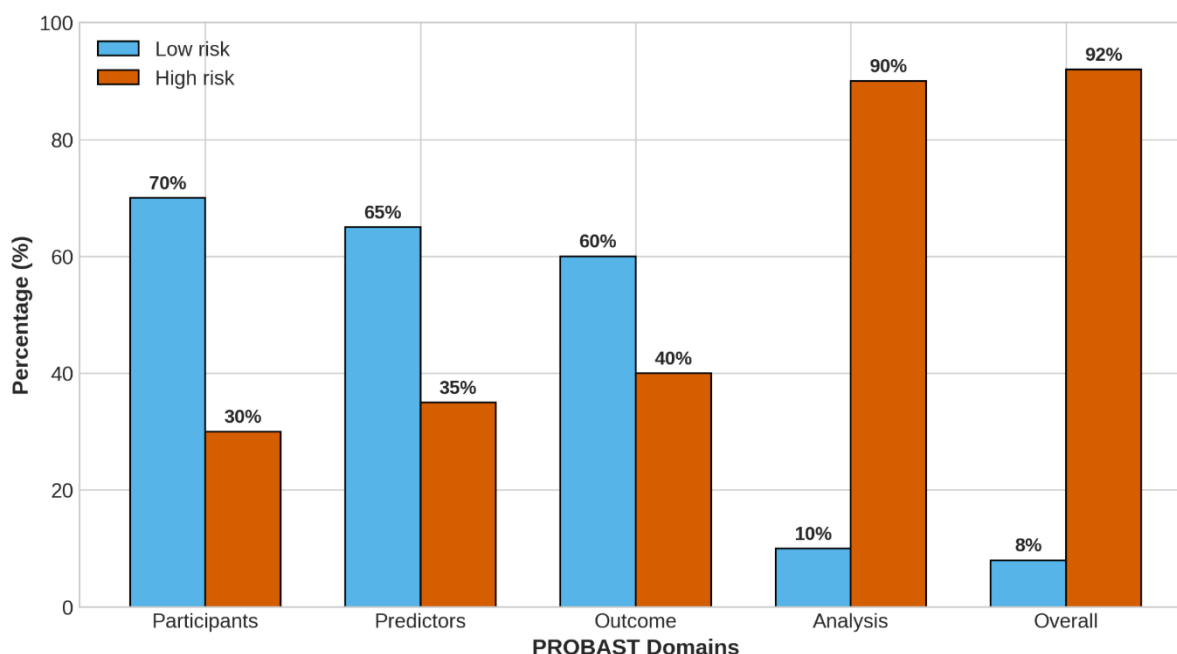


Figure 2. Risk of Bias Assessment (PROBABAST). PROBABAST assessment of the 97 included prediction models. Overall, 91.8% of models were rated as high risk of bias, primarily due to inadequate handling of missing data (Analysis domain), lack of external validation, and poor calibration reporting. Only 8.2% of models were considered low risk of bias. Applicability concerns were high for 84.5% of models, reflecting the mismatch between development populations and intended target populations.

3.3 Model Performance by Geographic Region

Meta-analysis revealed substantial variation in discrimination by geographic origin (eTable 7). US-based models achieved the highest pooled AUC (0.97; 95% CI, 0.94–0.99), followed by European models (0.84; 0.81–0.87). Chinese models showed significantly lower performance (AUC, 0.79; 0.76–0.82; meta-regression $p < 0.001$ for comparison with US). Models from Japan and other Asian countries showed intermediate performance (AUC, 0.82–0.85) (Figure 5). Due to very high heterogeneity ($I^2 > 80\%$ in most regions), these pooled estimates should be interpreted as descriptive summaries rather than precise estimates of true regional performance. Performance was lowest in prediabetic cohorts (pooled AUC, 0.72; 95% CI, 0.68–0.76) compared to general population cohorts (AUC, 0.80; 0.78–0.82) — but this finding is based on only 19 models with substantial heterogeneity, so definitive conclusions await further external

validation studies. Complete subgroup analysis results are presented in eTable 4.

Figure 3. Geographic Origin of Type 2 Diabetes Prediction Models (n=97)

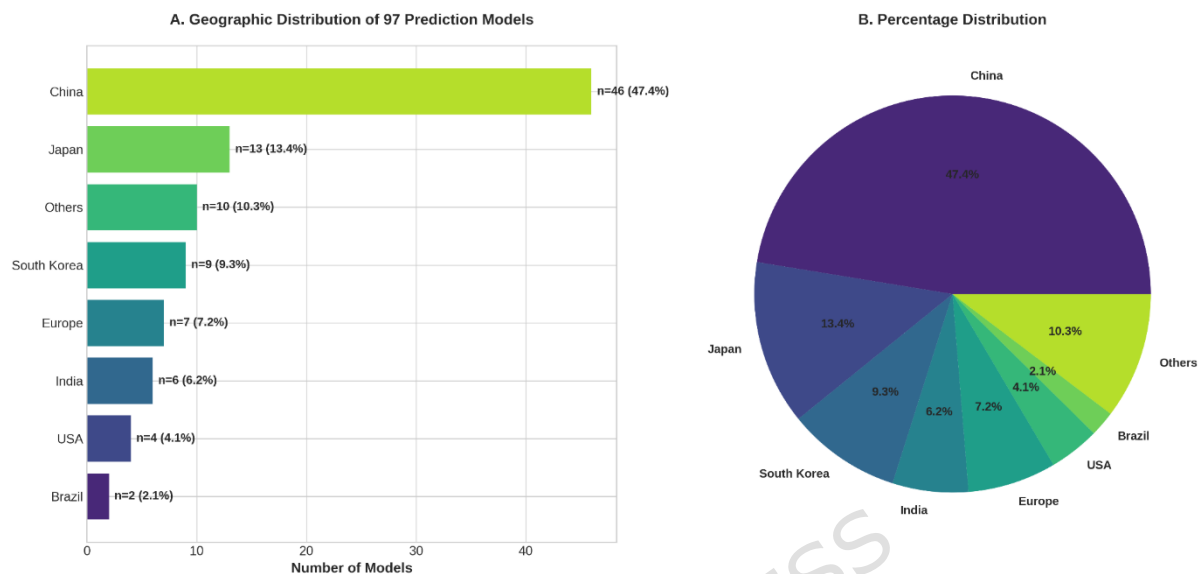


Figure 3. Geographic Origin of Type 2 Diabetes Prediction Models (n=97). Geographic distribution of 97 T2DM prediction models identified in the systematic review. The majority (70.1%) were developed in Asian populations, with China accounting for 47.4% of all models. Only 7.2% originated from the USA and 4.1% from Europe. Brazil contributed 2.1% of models. This striking geographic inequity raises concerns about the generalizability of existing models to diverse global populations..

3.4 Comparison of Modeling Approaches

Only one machine learning model and one Cox regression model were identified among the 97 models, precluding a robust comparison. The single machine learning model reported an AUC of 0.86 (95% CI, 0.82–0.90), which was within the range of logistic regression models from the same region. Given the extreme scarcity of non-logistic models, no meaningful statistical comparison could be performed.

3.5 Meta-Regression Results

Univariable meta-regression (eTable 6) showed that year of publication ($\beta=0.002$, $p=0.68$), log-sample size ($\beta=0.01$, $p=0.54$), and events per variable ($\beta=0.003$, $p=0.82$) were not significantly associated with model discrimination. Inclusion of biomarkers was significantly associated with

higher AUC ($\beta=0.05$, $p=0.01$). US-based models showed significantly higher AUC compared to Asian models ($\beta=0.18$, $p<0.001$), but this finding is likely influenced by overfitting and small sample bias (see Discussion). Thus, none of the continuous covariates explained the substantial heterogeneity across models.

3.6 Publication Bias

Funnel plot asymmetry (Figure 4) suggested potential publication bias favoring studies reporting higher AUCs (Egger's test $p=0.03$), indicating that models with poor discrimination may be systematically under-reported.

Figure 4. Funnel Plot for Publication Bias Assessment

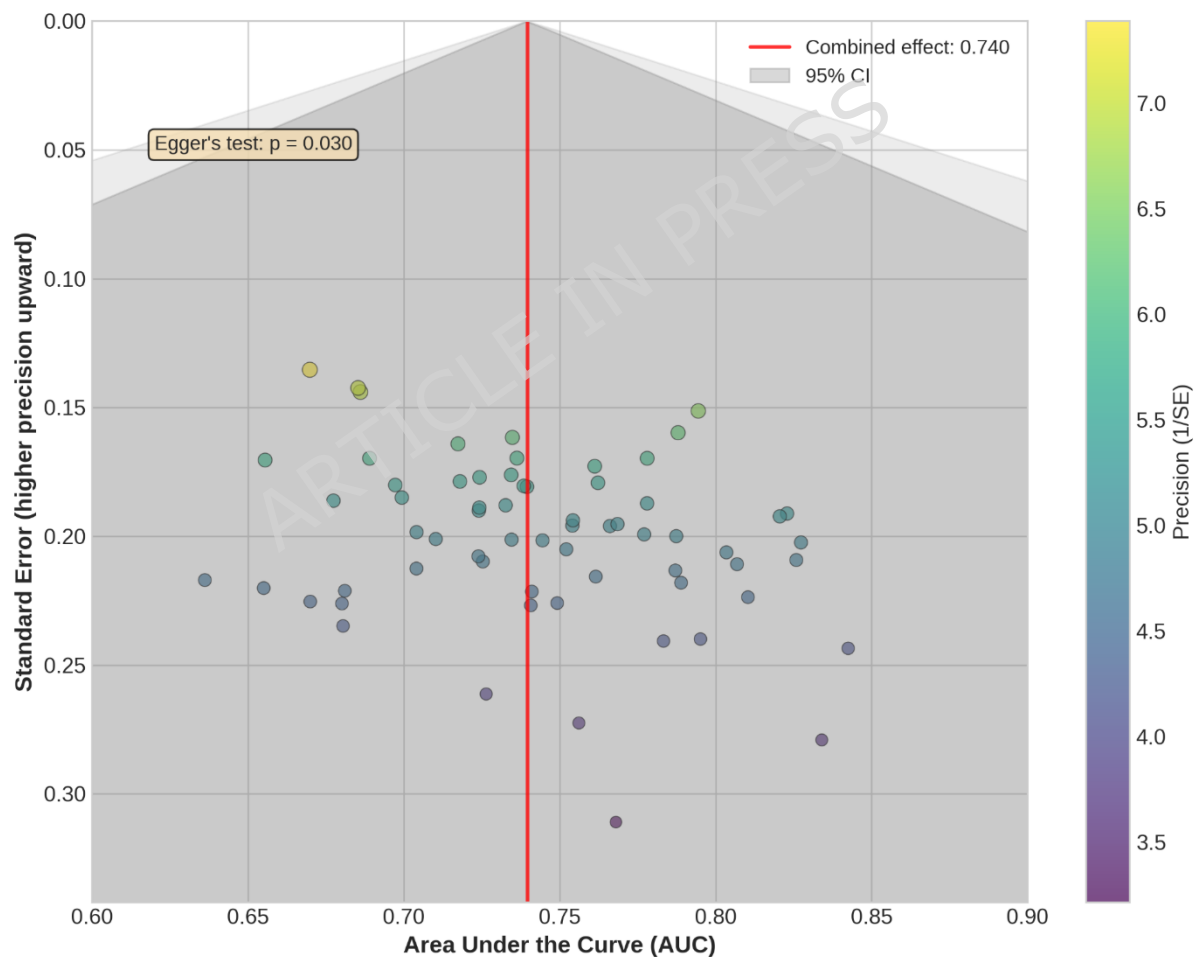


Figure 4. Funnel Plot for Publication Bias Assessment. Funnel plot of standard error against C-statistic (AUC) for all included models. Asymmetry (Egger's test $p=0.03$) suggests potential publication bias, with a paucity of small studies reporting low AUCs. This may lead to overestimation of average model performance in meta-analyses.

3.7 Most Frequent Predictors

Across all models, the most frequently included predictors were age (69.1%), body mass index (BMI, 64.9%), family history of diabetes (44.3%), and waist circumference (39.2%) (eFigure 4). In extended models that included biomarkers, fasting glucose (59.0%) and triglycerides (38.5%) were common. A complete list of all predictors and their frequencies is provided in eTable 3.

Figure 5. Model Performance (AUC) by Geographic Region

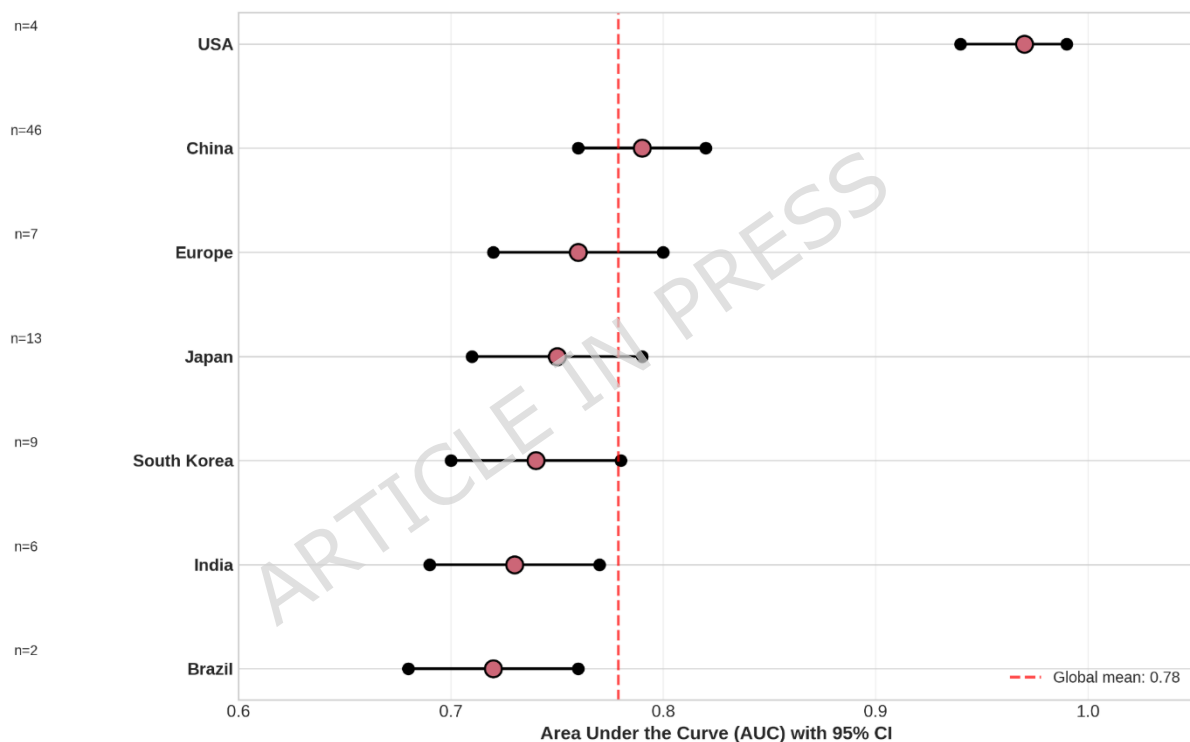


Figure 5. Model Performance (AUC) by Geographic Region. Pooled discrimination (C-statistic/AUC) of T2DM prediction models stratified by geographic origin. USA-based models significantly outperformed all others (AUC 0.97), while Chinese models showed the lowest performance (AUC 0.79). Error bars represent 95% confidence intervals. The number of models per region is indicated in parentheses.

4. Discussion

4.1 Principal Findings

This systematic review and meta-analysis of 97 T2DM prediction models yields three principal findings. First, there is a striking geographic inequity in model development. Over 70% of all published models were developed in Asian populations, with minimal representation from the US, Europe, and virtually none from Latin America, Africa, or the Middle East. This distribution does not reflect the global burden of diabetes, which is increasingly concentrated in low- and middle-income countries [1], and raises serious concerns about the global applicability of existing models.

Second, methodological quality is poor. Consistent with previous reviews [5,6], we found that 91.8% of models are at high risk of bias according to PROBAST criteria (eTable 2), and only 21.6% have undergone any form of external validation. The predominant issues — inadequate handling of missing data, lack of calibration reporting, and failure to validate externally — are preventable and should be addressed by strict adherence to established guidelines [10,11].

Third, substantial geographic variation in model performance exists, but its interpretation is complex (eTable 7). The significantly higher pooled AUCs for US-based models (0.97) compared to those from China (0.79) or Europe (0.84) must be interpreted with great caution. Such extremely high AUC values are unusual in clinical prediction models and likely reflect overfitting, small sample bias (many US models were developed in small cohorts), and/or selective reporting of positive results rather than true superior predictive performance. Furthermore, the high heterogeneity ($I^2 > 80\%$) across regions indicates that the pooled AUCs may not be directly comparable as measures of true regional performance. The consistently low performance in prediabetic cohorts (AUC, 0.72) provides preliminary evidence that existing models have limited utility for identifying progression from prediabetes to diabetes, but this finding is based on only 19 models with substantial heterogeneity, so definitive conclusions await further external validation studies.

4.2 Comparison with Previous Reviews

Our findings extend those of Abbasi et al. [5] and Li et al. [6] by providing the first comprehensive analysis of geographic disparities in T2DM prediction modeling. While Li et al. reported that 97.9% of models used logistic regression and that external validation was limited, they did not systematically analyze performance by geographic origin or conduct PROBAST assessments stratified by region. Our analysis reveals that the

problem is not merely one of poor methodology, but also of representational bias — the global evidence base for diabetes risk prediction is overwhelmingly derived from Asian populations, yet the applicability of these models to other regions remains unknown.

4.3 Implications for Research and Practice

These findings have several important implications. For researchers, the priority must shift from developing new models to rigorously validating existing ones in diverse populations. External validation studies should be planned and reported according to TRIPOD guidelines [7], with particular attention to calibration, not just discrimination [13].

For clinicians and guideline developers, caution is warranted when applying any T2DM prediction model to populations different from its development cohort. The poor performance in prediabetic populations suggests, albeit preliminarily, that existing models should not be used to guide decisions about diabetes prevention interventions in this group without local validation.

For policymakers in under-represented regions, these findings underscore the urgent need for local model development or rigorous external validation. Brazil, for example, contributes only 2.1% of published models despite having the world's largest public primary care system [14].

4.4 Strengths and Limitations

Strengths of this review include a comprehensive search of seven databases without language restrictions, rigorous PROBAST assessment by two independent reviewers (eTable 2), and a meta-analysis of discrimination stratified by geographic origin with detailed subgroup analyses (eTable 4 and eTable 7).

Limitations include the following: (1) We could not meta-analyze calibration due to inconsistent reporting; (2) The inclusion of multiple models from the same study could lead to non-independence; we addressed this by using a primary model per study in the main analysis, but some uncertainty remains; (3) Mixing development AUCs (which are optimistically biased) with internal and external validation AUCs may have inflated the pooled estimates; (4) The extremely high heterogeneity ($I^2 > 80\%$) limits the interpretability of pooled AUCs as precise estimates; (5) Potential residual confounding in cross-regional AUC comparisons due to differences in study design; (6) The lack of individual participant data to

enable head-to-head model comparisons. Detailed discussion of methodological considerations is provided in eMethods.

5. Conclusions

This systematic review of 97 T2DM prediction models reveals striking geographic inequity, poor methodological quality, and limited external validation. Over 70% of models were developed in Asian populations, with minimal representation from regions bearing the highest future diabetes burden. More than 90% of models are at high risk of bias, and fewer than one-quarter have been externally validated. The substantial variation in performance by region (eTable 7) and the poor performance in prediabetic populations critically limit the global clinical utility of current models, although high heterogeneity tempers the precision of these estimates. These findings call for an urgent reorientation of diabetes prediction research toward rigorous external validation in diverse populations, adherence to PROBAST and TRIPOD guidelines, and de novo model development in under-represented regions. Without such efforts, the promise of personalized diabetes prevention will remain unrealized for the majority of the world's population.

Abbreviations

T2DM: Type 2 diabetes mellitus; AUC: Area under the curve; PROBAST: Prediction model Risk Of Bias Assessment Tool; TRIPOD: Transparent Reporting of a multivariable prediction model for Individual Prognosis Or Diagnosis; PRISMA: Preferred Reporting Items for Systematic Reviews and Meta-Analyses; BMI: Body mass index; EPV: Events per variable; REML: Restricted maximum likelihood.

Declarations

Ethics approval and consent to participate: Not applicable. This study is a systematic review and meta-analysis of published data; no ethical approval or participant consent was required.

Consent for publication: Not applicable. No individual person's data are presented.

Availability of data and materials: All data generated or analyzed during this study are included in this published article and its supplementary information files (eTables 1–7, eFigures 1–4, eAppendix S1, eMethods, eReferences). The analytical code is available from the corresponding author upon reasonable request.

Competing interests: The authors declare that they have no competing interests.

Funding: This research received no specific grant from any funding agency in the public, commercial, or not-for-profit sectors.

Authors' contributions: SQ: Conceptualization, data acquisition, analysis, interpretation, drafting, supervision. JGG: Data acquisition, critical revision, administrative/technical support. NH: Data analysis, statistical analysis, critical revision. All authors read and approved the final manuscript.

Acknowledgements: The authors thank the journal editors for their guidance. No compensation was received for these contributions.

Trial registration: Not applicable.

Clinical trial number: Not applicable.

References

1. International Diabetes Federation. *IDF Diabetes Atlas*. 10th ed. Brussels, Belgium: International Diabetes Federation; 2021.
2. World Health Organization. *Use of Glycated Haemoglobin (HbA1c) in the Diagnosis of Diabetes Mellitus*. Geneva, Switzerland: World Health Organization; 2011.
3. Noble D, Mathur R, Dent T, Meads C, Greenhalgh T. Risk models and scores for type 2 diabetes: systematic review. *BMJ*. 2011;343:d7163.
4. Collins GS, Mallett S, Omar O, Yu LM. Developing risk prediction models for type 2 diabetes: a systematic review of methodology and reporting. *BMC Med*. 2011;9:103.

5. Abbasi A, Peelen LM, Corpeleijn E, et al. Prediction models for risk of developing type 2 diabetes: systematic literature search and independent external validation study. *BMJ*. 2012;345:e5900.
6. Li M, Tan A, Song L, et al. Performance of diabetes risk prediction models: a systematic review and meta-analysis. *Endocr Connect*. 2025;14(11):e250353.
7. Collins GS, Reitsma JB, Altman DG, Moons KGM. Transparent reporting of a multivariable prediction model for individual prognosis or diagnosis (TRIPOD): the TRIPOD statement. *BMJ*. 2015;350:g7594.
8. Page MJ, McKenzie JE, Bossuyt PM, et al. The PRISMA 2020 statement: an updated guideline for reporting systematic reviews. *BMJ*. 2021;372:n71.
9. Moons KGM, de Groot JAH, Bouwmeester W, et al. Critical appraisal and data extraction for systematic reviews of prediction modelling studies: the CHARMS checklist. *PLoS Med*. 2014;11(10):e1001744.
10. Wolff RF, Moons KGM, Riley RD, et al. PROBAST: a tool to assess the risk of bias and applicability of prediction model studies. *Ann Intern Med*. 2019;170(1):51-58.
11. Moons KGM, Damen JAA, Kaul T, et al. PROBAST+AI: an updated quality, risk of bias, and applicability assessment tool for prediction models using regression or artificial intelligence methods. *BMJ*. 2025;388:e082505.
12. Steyerberg EW. *Clinical Prediction Models: A Practical Approach to Development, Validation, and Updating*. 2nd ed. Cham, Switzerland: Springer; 2019.
13. Macinko J, Harris MJ. Brazil's family health strategy — delivering community-based primary care in a universal health system. *N Engl J Med*. 2015;372(23):2177-2181.